

The immunobiology of mushrooms.

There has been enormous interest in the biologic activity of mushrooms and innumerable claims have been made that mushrooms have beneficial effects on immune function with subsequent implications for inhibition of tumor growth. The majority of these observations are anecdotal and often lack standardization. However, there remains considerable data on both in vitro and in vivo effects that reflect on the potential of mushroom compounds to influence human immunity. A number of these effects are beneficial but, unfortunately, many responses are still characterized based on phenomenology and there is more speculation than substance. With respect to tumor biology, although many neoplastic lesions are immunogenic, tumor antigens frequently are self antigens and induce tolerance and many patients with cancer exhibit suppressed immune responses, including defective antigen presentation. Therefore, if and when mushroom extracts are effective, they more likely function as a result of improved antigen presentation by dendritic cells than by a direct cytopathic effect. In this review we attempt to place these data in perspective, with a particular focus on dendritic cell populations and the ability of mushroom extracts to modulate immunity. There is, at present, no scientific basis for the use of either mushrooms or mushroom extracts in the treatment of human patients but there is significant potential for rigorous research to understand the potential of mushrooms in human disease and thence to focus on appropriate clinical trials to demonstrate effectiveness and/ or potential toxicity.